

Question Paper

Exam Date & Time: 28-Dec-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (INFORMATION ECHNOLOGY) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ICT 2124/ICT_2124 - PRINCIPLES OF DATA COMMUNICATION

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Consider a transmitter that operates at a power level of 20 W. Convert this transmit power into units of dBm and dBW. If the transmitter's power is applied to a unity gain antenna at a frequency of 2.4 GHz, calculate the received power in dBm at a free space distance of 150 meters. Repeat this calculation for a distance of 5 kilometers. Finally, recalculate the received power at 5 kilometers, assuming a receiver antenna gain of 3 dB. Analyze how variations in receiver gain affect the overall signal strength and reliability of the communication link. (5)
- 1B) Ten sources, each with a bit rate of 300 kbps, are to be combined using multilevel time-division multiplexing with 5 synchronizing bits added to each frame. Answer the following questions about the final stage of the multiplexing: (3)
- i) Calculate the size of a frame in bits?
 - ii) Calculate the frame rate?
 - iii) Calculate the duration of a frame?
 - iv) Calculate the data rate?
- 1C) How does a dual ring topology maintain network integrity during normal operations and in a failure scenario? Use a diagram to show both conditions. (2)
- 2A) An 8-bit data word is given as 11011010. Generate the Hamming code for this data word using even parity. Suppose the receiver receives the codeword as 110011010101. Identify the error in the received codeword and correct it. Show the steps involved in generating the Hamming code, explain how the error position is determined, and correct the codeword. (5)
- 2B) A communication channel operates in the frequency range of 200 MHz to 215 MHz with an SNR of 18 dB. What is the Shannon limit on the capacity of this channel? What would be the impact on the capacity if the SNR were reduced to 10 dB? (3)
- 2C) Analyze how the choice of modulation technique in an FDM system can influence the quality of the transmitted signals. Provide an example of a modulation technique that can improve signal integrity. (2)
- 3A) Depict the communication between a transmitter (A) and a receiver (B) using the HDLC protocol with selective reject ARQ, assuming a window size of 6. The following events take place: (5)
- i) A initializes the link, and B acknowledges the initialization.
 - ii) B transmits the first six data frames (Frame 1 to Frame 6).
 - iii) Frame 2 is damaged during transmission, and A sends a reject (REJ) for Frame 2.
 - iv) The REJ frame sent by A is lost, so B does not know about the rejection.
 - v) B continues to transmit frames 3, 4, 5, and 6, unaware of the lost REJ.
 - vi) After not receiving acknowledgment for Frame 2, B experiences a timeout and retransmits Frame 2.
 - vii) A acknowledges the successful reception of Frame 2 after its retransmission.
 - viii) B sends a disconnect request, and A acknowledges the disconnection.
- 3B) Describe a scenario in which this lack of synchronization could create timing errors. Illustrate the possible effects of these timing errors on the clarity and accuracy of the received information. (3)

- 3C) In a digital communication system with an available bandwidth of 250 kHz, spanning from 600 kHz to 850 kHz, determine the carrier frequency and bit rate for a Quadrature Phase Shift Keying modulation scheme, given a roll-off factor $r=0.25$? (2)
- 4A) In a network consisting of five LANs (LAN A, LAN B, LAN C, LAN D, LAN E) connected by four bridges (B1, B2, B3, and B4), the administrator has configured the Spanning Tree Protocol (STP) to prevent loops. Due to a network upgrade, redundant links were added, which led to some bridges being excluded from the spanning tree. (5)
- i) Explain how the Spanning Tree Protocol detects and prevents loops in a bridged network.
 - ii) Show how the spanning tree structure will change after a failure of bridge B1, and describe how STP reconfigures the network.
 - iii) Identify which bridges and links are redundant and explain how redundancy improves network reliability.
- 4B) A data stream "1100101" is encoded using Manchester encoding. Calculate the encoded bit sequence. (3)
- 4C) Analyze how increasing the number of subcarriers affects the system's bandwidth and data rate in an OFDM system, considering both advantages and limitations. (2)
- 5A) A Slotted ALOHA system with 200-bit frames operates with a bandwidth of 250 kbps. The network currently generates 600 frames per second. (5)
- i) Calculate the current throughput.
 - ii) If the frame generation rate increases to 900 frames per second, recalculate the throughput. Explain the effect of the increased frame generation rate on collisions and throughput.
- 5B) In a Stop-and-Wait ARQ system, the round-trip time (RTT) between the sender and the receiver is 30 ms, and the data rate is 1 Mbps. Each frame contains 1000 bits. (3)
- i) Calculate the efficiency of the Stop-and-Wait ARQ protocol.
 - ii) Analyze the factors that affect the protocols efficiency and discuss how increasing the frame size or reducing the RTT could improve the system's performance.
- 5C) A communication system is transmitting data at a rate of 12 Mbps using 16-QAM (16-level Quadrature Amplitude Modulation). Calculate the minimum bandwidth required using the Nyquist theorem. (2)

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